

13/10/25

Task 11: CRUD operations in Graph Databank

AIM:

To perform, Create, Read, Update and Delete operations on a Graph Databank for the Hotel Management System

Steps and cypher commands

1- Create Nodes with Properties

```
CREATE (h:Hotel {Hotel_ID:"H001", Hotel_Name:"Surya Inn",  
City:"Chennai"});
```

```
CREATE (g:Guest {Guest_ID:"G101", Guest_Name:"John", Phone:"9876543210"});
```

```
CREATE (r:Room {Room_No:"101", Room_Type:"Deluxe"});
```

Output

Added 3 nodes successfully with their properties

2- View All Nodes

```
MATCH (n) RETURN n;
```

Output:

Displays all Hotel, Guest and Room nodes

3- Create Relationships

```
MATCH (g:Guest {Guest_Name:"John"}, (r:Room {Room_No:"101"})
```

```
CREATE (g)-[BOOKED]->(r)
```

```
RETURN g,r;
```

```
MATCH (r:Room {Room_No:"101"}, (h:Hotel {Hotel_Name:"Surya Inn"})
```

```
CREATE (r)-[:ASSIGNED-TO]->(h)
```

```
RETURN r,h;
```

Output:

Relationships BOOKED and ASSIGNED-TO created successfully

OUTPUT:

match(n) return(n)

John

Mark

Summit Inn

Ben

match(n: Hotel) return(n)

John

Mark

Ben

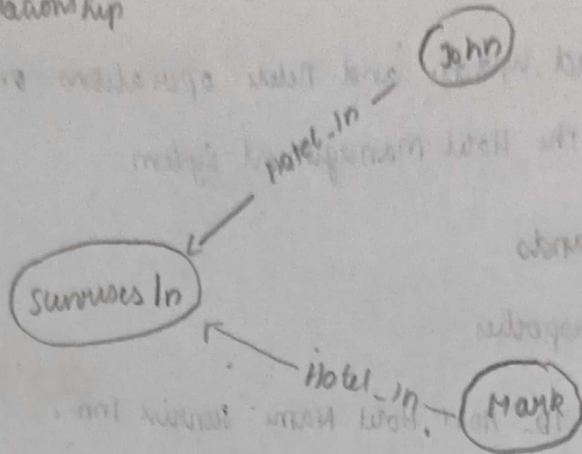
Relationship

Summit Inn

Hotel

John

Relationship



4- Retrieve Relationship Data

```
MATCH(g:Guest)-[:BOOKED]->(r:Room)-[:ASSIGNED_TO]->(h:Hotel)
RETURN g.Guest_Name, r.Room_No, h.Hotel_Name;
```

Output:

Displays Guest name, Room number and Hotel name

5- Delete a Node

```
MATCH(g:Guest {Guest_Name:"John"}) DETACH DELETE g;
```

Output:

Deleted one node and its connected relationships

VEL TECH - CSE	
EX NO.	11
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	15
SIGN WITH DATE	13/10/20

RESULT:

Thus the CRUD operations in graph database for Hotel management system executed successfully